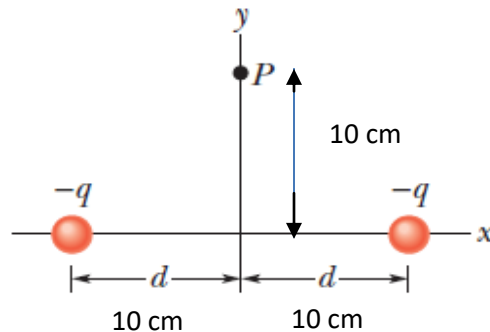


Practice Problem Sheet-4

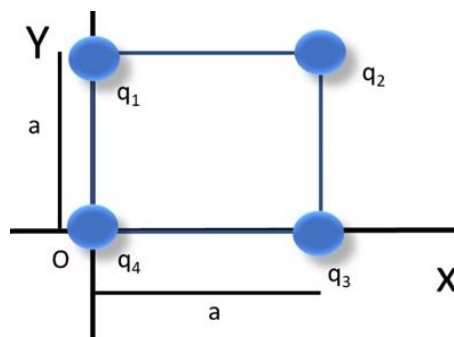
Course Code: PHY 2105/PHY 105 Summer 2026 [262]

Course Title: Physics Content: Electric Field

- Two equal charges $q=12\times 10^{-7}\text{C}$ are placed at the two corners of an equilateral triangle of side $r=10\text{cm}$. Draw the triangle with charges. Find the resultant electric field and its direction at third corner of the triangle.
- Two point charges $+4q\text{ C}$ and $+q\text{ C}$ are placed 30 cm apart. At what point on the line joining them the electric field is zero?
- Two equal charges of $10\times 10^{-5}\text{ C}$ are shown in fig below; each produces an electric field at point P on Y axis. (i) What is the magnitude of the electric field of each charge at P ? (ii) What is direction of each field? (iii) What is the magnitude of the net field at P ? (iv) What is the direction of the net field? and (v) Find the X and Y components of the field vector.

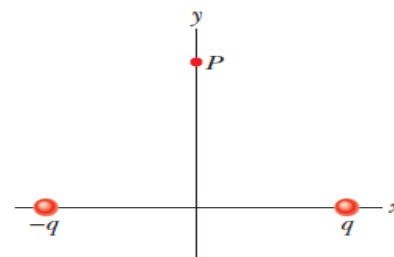


- A charged particle produces an electric field with a magnitude of 5.0 N/C at a point that is 60 cm away from the particle. What is the magnitude of the particle's charge?
- In the figure, the four particles form a square of edge length $a=5.00\text{ cm}$ and have charges $q_1 = +10.00\text{ nC}$, $q_2 = -20.0\text{ nC}$, $q_3 = +20.0\text{ nC}$ and $q_4 = -10.0\text{ nC}$. In unit vector notation, what net electric field does the particle produce at the square's center? What is the net field direction?



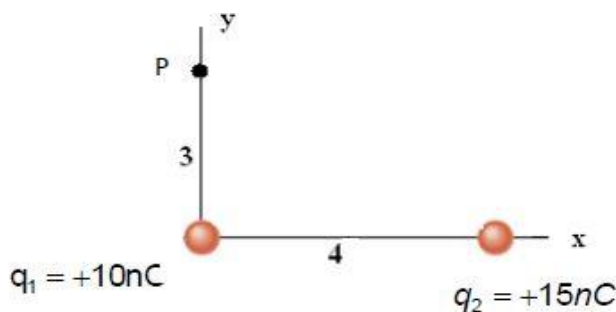
- Four charges $+2q$, $+4q$, $+2q$ and $-2q$ are placed at the corners of a square. (i) Draw the arrangement of the charges. (ii) Calculate the magnitude and direction of electric field at the intersection of the diagonals of the square of side 10 cm if $q=57\times 10^{-9}\text{C}$.
- In figure, two charged particles on an x-axis: $-q = -4.8\times 10^{-19}\text{ C}$ is at $x = -3.00\text{ m}$ and $q = 4.80\times 10^{-19}\text{ C}$ is at $x = +3.00\text{ m}$. What are the (i) magnitude and (ii) direction of the net electric field produced at

point P in the Y axis at $y = 4.00$ m. (iii) if $-q=0$ C and at O position (0,0) $+2q$ charge is inserted then

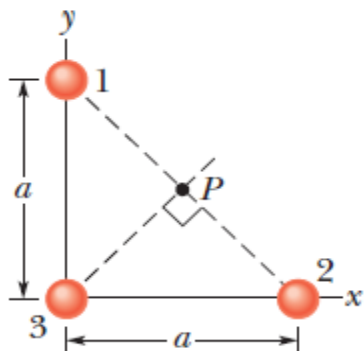


also find out the direction and magnitude of the net electric field.

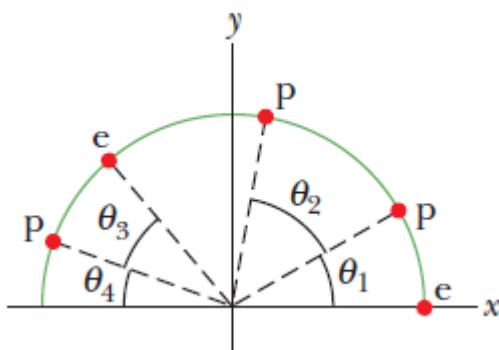
8. Calculate the Electric Field due to a proton at the location of the electron in the H atom. The radius of the electron orbit is 0.5×10^{-10} m.
9. A charge of $-1.0 \mu\text{C}$ is located at the coordinates (0,2) while a second charge of $+1.0 \mu\text{C}$ is located at the coordinates (1,0). Draw the charge arrangement and determine the value of the following quantities at the origin: (i) the magnitude of the electric field E and (ii) the direction of the electric field.
10. An electric dipole consists of charge $+2e$ and $-2e$ separated by 0.75nm . It is in an electric field of strength $5.4 \times 10^{-8} \text{ N/C}$. (i) Calculate the magnitude of the torque on the dipole when the dipole moment is (a) parallel to and (b) perpendicular to the electric field. (ii) How much work is required to turn an electric dipole 180° in a uniform electric field of magnitude $E = 5.4 \times 10^{-8} \text{ N/C}$ if the dipole moment has the initial angle 65° .
11. A neutral salt molecule (NaCl) in its aqueous state has an electric dipole moment of magnitude $5 \times 10^{-32} \text{ Cm}$. (i) How far apart are the molecule's centers of positive and negative charge? (ii) If the molecule is placed in an electric field of $2.2 \times 10^5 \text{ N/C}$, what maximum torque can the field exert on it? (iii) How much work must an external agent do to rotate this molecule by 170° in this field, starting from its fully aligned position, for which $\theta = 0^\circ$? (iv) What is the maximum potential energy?
12. A charge of $q_1 = +10 \text{ nC}$ is located at the origin. While a second charge of $q_2 = +15 \text{ nC}$ is located at the coordinates (4,0). Draw the charge arrangement and determine the value of the following quantities at the coordinates (0,3) i.e., $y=3\text{m}$ and $x=0$ m, (i) the magnitude of the net electric field E and (ii) the direction of the net electric field.



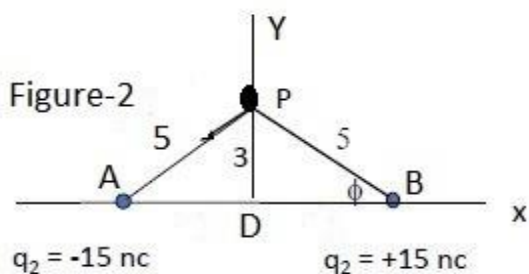
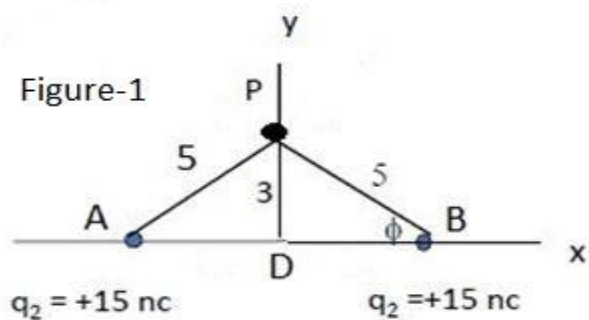
13. The three particles are fixed in place and have charges $q_1 = q_2 = +e$ and $q_3 = +2e$. Distance $a = 6.00 \mu\text{m}$. What are the (i) magnitude and (ii) direction of the net electric field at point P due to the particles?



14. An electric dipole with dipole moment $\vec{P} = (3\hat{i} + 4\hat{j})(1.24 \times 10^{-30}) \text{ Cm}$ is found in an electric field $\vec{E} = (4000 \text{ N/C}) \hat{i}$ (i) What is the potential energy of the electric dipole? (ii) What is the torque acting on it? (iii) If an external agent turns the dipole until its electric dipole moment is $\vec{P} = (-4\hat{i} + 3\hat{j})(1.24 \times 10^{-30}) \text{ Cm}$, how much work is done by the agent?
15. Figure shows an uneven arrangement of electrons (e) and protons (p) on a circular arc of radius $r = 2.00 \text{ cm}$, with angles $\theta_1 = 30^\circ$, $\theta_2 = 50^\circ$, $\theta_3 = 30^\circ$, and $\theta_4 = 20^\circ$. What are the (i) magnitude and (ii) direction (relative to the positive direction of the x axis) of the net electric field produced at center of the arc?

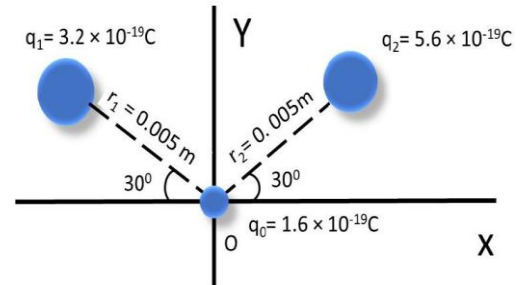


16. ΔPAD and ΔPBD are two right angles triangle. (i) What is the field on the y axis at point P for Figure-1? and (ii) What is the field on the y axis at point P for Figure-2?



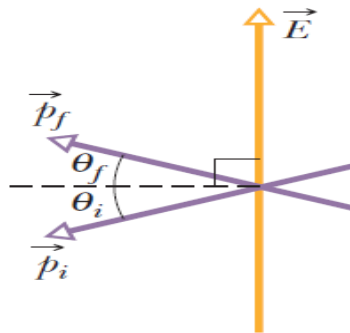
17. An electron enters a region of uniform electric field with an initial velocity of 40 km/s in the same direction as the electric field, which has magnitude $E = 50 \text{ N/C}$. (i) What is the speed of the electron 1.5 ns after entering this region? (ii) How far does the electron travel during 1.5 ns the interval?
18. An electron is projected perpendicularly to a downward electric field of $E = 2000 \text{ N/C}$ with a horizontal velocity $v = 10^6 \text{ m/s}$. How much is the electron deflected after traveling 1 cm ? Given, $e = 1.6 \times 10^{-19} \text{ C}$ and $m_e = 9.1 \times 10^{-31} \text{ kg}$.

19. An electron on the Y axis is 25nm from the center of a charged ring. What is the magnitude of the electric field and electrostatic force on the electron if the ring is staying at a large distance? Assume that 1nm is the radius of the ring which is much smaller than the separation of the charged particle e^- from the center.
20. From the figure below, (i) Calculate the magnitude of the net electric field on the point O if $q_0=0$. (ii)

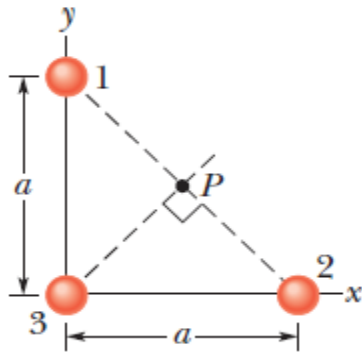


Calculate the direction of net field on the point O if $q_0=0$.

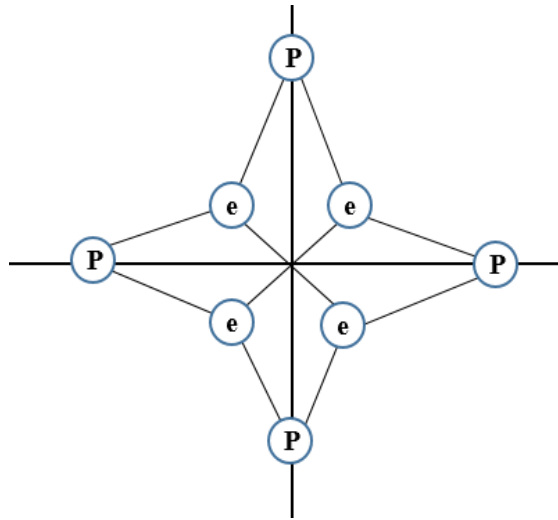
21. A charged cloud system produces an electric field in the air near Earth's surface. A particle of charge $-2.0 \times 10^{-9} \text{ C}$ is acted on by a downward electrostatic force of $3.0 \times 10^{-6} \text{ N}$ when placed in this field. (i) What is the magnitude of the electric field? What are the (ii) magnitude and (iii) direction of the electrostatic Force $\vec{F}_{el}$ on the proton placed in this field? (iv) What is the magnitude of the gravity force $\vec{F}_g$ on the proton? (v) What is the ratio $\vec{F}_{el}/\vec{F}_g$ in this case?
22. In Figure below, an electric dipole produced in a water molecule with dipole distance 1.2 fm swings from an initial orientation i ($\theta_i = 20.0^\circ$) to a final orientation f ($\theta_f = 20.0^\circ$) in a uniform external electric field $\vec{E}$. The electric dipole moment is $1.60 \times 10^{-27} \text{ Cm}$ and the field magnitude is $3 \times 10^6 \text{ N/C}$. (i) What is the change in the dipole's potential energy? (ii) What is the electric field if the path length is equal to the dipole distance?



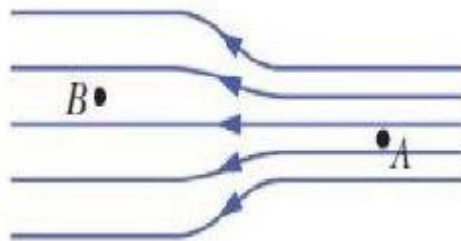
23. Fair weather atmospheric electricity is acting downward 100 km high in the ionosphere. The ionosphere voltage is $3 \times 10^6 \text{ V}$. What is the electric field required for producing high voltage?
24. The three particles are fixed in place and have charges $q_1 = q_2 = 5 \text{ nC}$ and $q_3 = 2 \text{ nC}$. If, distance $a = 5 \text{ nm}$ and $q_3 = 0 \text{ nC}$, what are the (i) magnitude and (ii) direction of the net electric field at point 3 due to the particles?



25. From the figure below, calculate the net electric field required to bring a $+2.2\mu\text{C}$ charge in the center of the shape. The distance of the electrons from the center is 0.1 cm and the distance of proton is 0.6 cm .



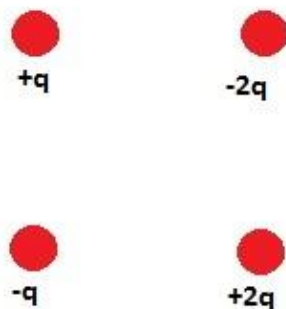
26. In the figure, the electric field lines on the left have twice the separation of those on the right. (i) If the magnitude of the field at A is 40 N/C , what is the magnitude of the force on a proton at A? (ii) What is



the magnitude of the field at B?

27. Two charges ($q_1 = +7\mu\text{C}$ and q_2) are located on the x axis on a coordinate system. They are both positive, but the second charge has twice the magnitude of the first. q_1 is at -0.5 m while q_2 is at $+0.5\text{ m}$. What are the magnitude and direction of net force at origin?
28. Four equal charges q are located on the corners of a square. What is the magnitude of the E-field at the center position of the square?

29. What are the magnitude and direction of the net E-field at the center of the square charge distribution shown below? Note that the charges are NOT equal. Assume that the sides of the square have a length



L.

30. A charge of $20 \mu\text{C}$ is 20 cm from another charge of $60 \mu\text{C}$. (i) Find the force between them. (ii) Determine the E field midway between them, and (iii) How much force will be experienced by an electron placed midway between them.
31. At the three consecutive corners of a square, 10 cm on each side, are point charges $4 \times 10^{-6} \text{ C}$, $8 \times 10^{-6} \text{ C}$, and $12 \times 10^{-6} \text{ C}$, respectively. Find the net electric field at the fourth corner of the square.
32. Three particles with charges $q_1 = 2Q$, $q_2 = -2Q$, and $q_3 = 4Q$ are placed at the same distance d from a test charge placed at the origin. Charges q_1 , q_2 , and q_3 and point A are arranged as you see below. Calculate the electric field produced at the point A in terms of k , Q , and d .

